

NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES

SCHOOL OF BUSINESS MANAGEMENT



MBA Core

Trimester IV

MULTIVARIATE DATA ANALYSIS

From Patterns to Predictions: Leveraging Clustering and LDA for Strategic Insights

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Introduction

Today, in an age of "big data," organizations in virtually every industry are relying on sophisticated statistical approaches to help them sift through extensive datasets to identify meaningful patterns. In this project, we show how Multivariate Data Analysis methods - Clustering and Linear Discriminant Analysis (LDA) - can be used to solve two separates, but related, business problems.

The first dataset we focus on is credit card customer data, where the goal is to group customers into meaningful groups based on their spending behaviours and account characteristics. By identifying natural clusters based on spending, businesses can design marketing campaigns, analyse product offerings, and develop retention strategies to focus on one of the identified customer segments like "high-value" customers.

The second dataset is about employee attrition, where the goal is to create a predictive classification model using LDA to differentiate employees who will stay from those that will leave. In doing so, human resource teams can better identify employees to prioritize retention efforts, allocate resources effectively, and reduce the costs of employee turnover.

Together, these analyses show how multivariate techniques can be leveraged in both exploratory (unsupervised) and predictive (supervised) contexts, providing insights that directly inform strategic decisions for the organization.

Objective

Objective overall:

Utilize multivariate analysis techniques on real business data so as to detect patterns, categorize observations and generate actionable insights for decision-makers.

Dataset 1 - Credit Card Customer Segmentation (Clustering)

- Identify independent customer segments based on transaction and account characteristics.
- Study the behavioural differences between the segments to develop engagement and marketing plans to reach the specific segments.
- Determine the appropriate number of clusters (4 vs. 5) and provide a rationale for the segmentation.
- Devise marketing strategies for each customer segment identified.

Dataset 2 - Employee Attrition Predictive Modelling (LDA)

- Develop a discriminant model that classifies employees as 'Likely to Stay' or 'Likely to Leave'.
- Assess which variables most distinguish between the two groups and quantify their relative importance.
- Develop actionable HR recommendations recognizing the likelihood of attrition in an employee and impact the level of workforce stability.

Cluster Analysis

About the Dataset

The sample Dataset summarizes the usage behaviour of about **9000 active credit card holders during the last 6 months**. The file is at a customer level with 18 behavioural variables.

Data Source: [Kaggle](#)

Dataset Variables:

CUST_ID	Identification of Credit Card holder (Categorical)
BALANCE	Balance amount left in their account to make purchases
BALANCE_FREQUENCY	How frequently the Balance is updated, score between 0 and 1 (1 = frequently updated, 0 = not frequently updated)
PURCHASES	Amount of purchases made from account
ONEOFF_PURCHASES	Maximum purchase amount done in one-go
INSTALLMENTS_PURCHASES	Amount of purchase done in instalment
CASH_ADVANCE	Cash in advance given by the user
PURCHASES_FREQUENCY	How frequently the Purchases are being made, score between 0 and 1 (1 = frequently purchased, 0 = not frequently purchased)
ONEOFFPURCHASESFREQUENCY	How frequently Purchases are happening in one-go (1 = frequently purchased, 0 = not frequently purchased)
PURCHASESINSTALLMENTSFREQUENCY	How frequently purchases in instalments are being done (1 = frequently done, 0 = not frequently done)
CASHADVANCEFREQUENCY	How frequently the cash in advance being paid
CASHADVANCECTR	Number of Transactions made with "Cash in Advanced"

PURCHASES_TRX	Number of purchase transactions made
CREDIT_LIMIT	Limit of Credit Card for user
PAYMENTS	Amount of Payment done by user
MINIMUM_PAYMENTS	Minimum amount of payments made by user
PRCFULLPAYMENT	Percent of full payment paid by user
TENURE	Tenure of credit card service for user

Problem Statement

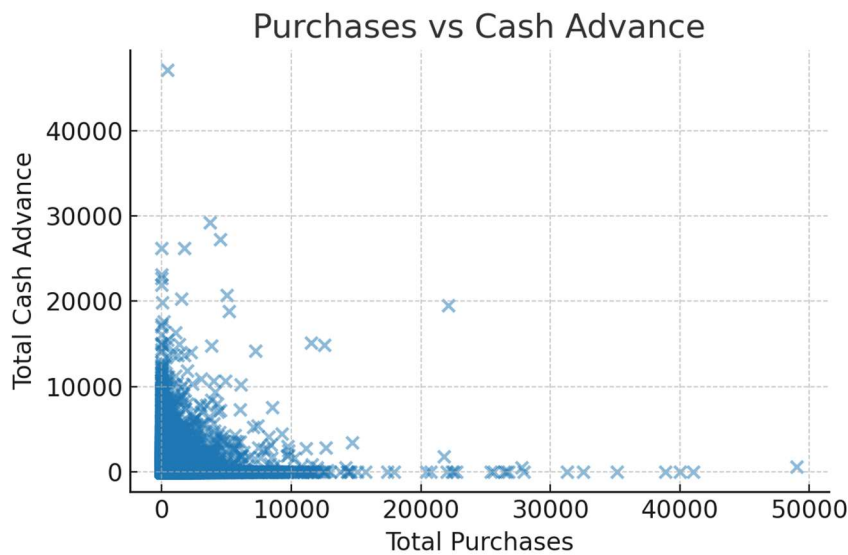
To identify customer segments of the credit card users and then devise marketing strategies for each type of customer segment so as to have focused marketing.

Link to working files:

https://drive.google.com/file/d/1P6YtzxFwhRAIhNJexEqS0fC3RINOWX3L/view?usp=drive_link

Visualisation

Purchases vs Cash Advances - Scatter Plot



Observations:

- Most of the customers have relatively low values for both Purchases and Cash Advances (bottom-left cluster).
- There is a distinct inverse relationship: those customers with high Purchases are likely to have low Cash Advances, and those with high Cash Advances are likely to have low Purchases.
- There are very few extreme outliers in either direction - some customers will take very large cash advances, some will make very large purchases, and some rare customers have high amounts in both axes.
- The distribution is very biased towards low-spending, low-cash-advancing customers.

Implications for Clustering:

The scatter plot shows at least three behavioural segments:

- Low Activity Users: Low purchase volumes paired with low sums of cash advances (probably a dormant, or low willingness to connect account).
- Purchase Users: High purchase volumes, cash advances are low (probably using cards for retail type transactions).
- Cash Users: High cash advances, low purchases (probably using cards for liquidity needs).
- The pattern confirms that PURCHASES and CASH_ADVANCE are the primary discriminating variables; their presence indicates customer potential segmentation.

Business Relevance:

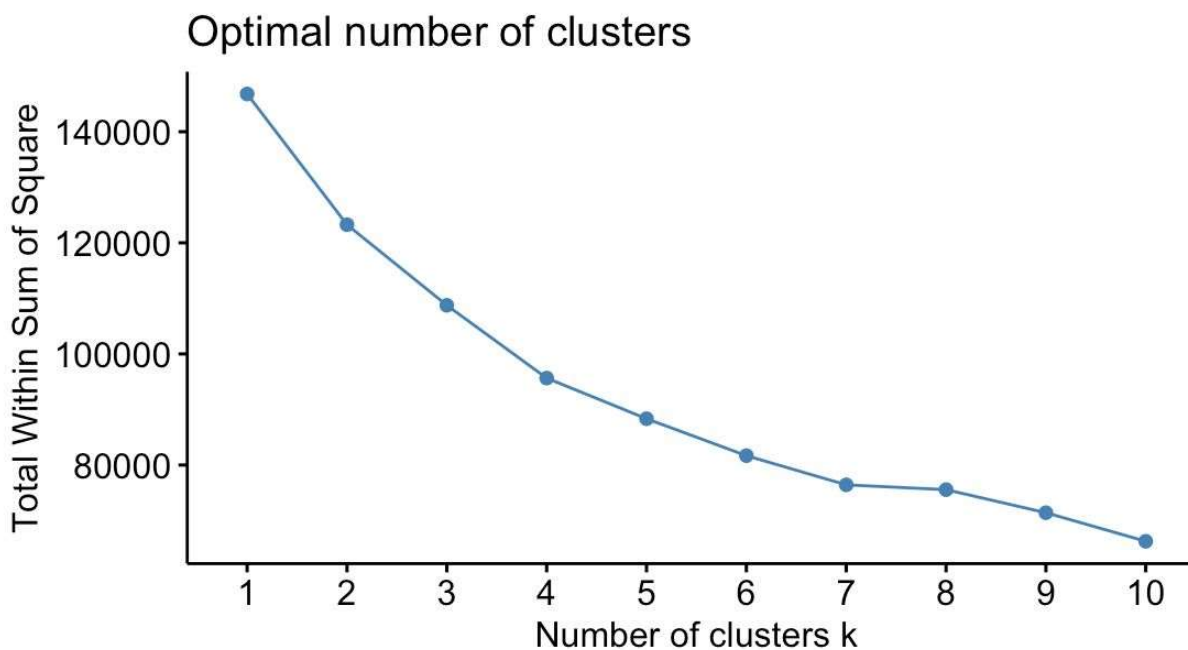
- Reward programs or exclusive offers may appeal to purchase-heavy customers.
- Cash-heavy customers may represent higher risk (i.e., credit stress) and may need interest rate modifications or credit limit reviews.
- Low activity customers can be re-engaged through marketing campaigns.

Methodology

K-means clustering is ideal for the given dataset which has all numeric values against the categorical column of 'Customer ID'.

The dataset was first cleaned to remove blank datapoints and further normalised before proceeding to perform k-means cluster analysis.

Elbow Plot:



Given no sharp elbow clustering was done taking $k=4$ & $k=5$ and results were compared.

- **X-axis (Number of clusters k):** Ranges from 1 to 10.
- **Y-axis (Total Within Sum of Squares / WSS):** Measures how tightly grouped the data points are within each cluster lower WSS means better clustering fit.
- **Trend:** As k increases, the WSS decreases because more clusters naturally fit the data better.
- **Elbow Point:** The rate of WSS reduction slows significantly around $k = 4$.
 - Before $k=4$, adding more clusters results in large drops in WSS.
 - After $k=4$, the improvement is marginal, suggesting diminishing returns.

Interpretation:

The "elbow" in the curve , where it starts bending is at $k \approx 4$, indicating that **4 clusters** is likely the optimal choice. This balances good cluster separation with model simplicity.

Code used on R for generating elbow plot

```
getwd()
setwd("~/Downloads/DivA_Grp7_cluster working files/csv & excel files/")
ccdata=read.csv("CC GENERAL cluster.csv",header=T)
ccdata
cleanccdata=na.omit(ccdata)
attach(cleanccdata)
View(cleanccdata)
write.csv(file="cleanccdata.csv",cleanccdata)
nor=scale(cleanccdata[,c(2:18)],center=TRUE,scale=TRUE)
nor
install.packages("factoextra")
library(factoextra)
fviz_nbclust(nor,kmeans,method="wss")
```

Link to working files:

https://drive.google.com/file/d/1P6YtzxFwhRAIhNJexEqS0fC3RINOWX3L/view?usp=drive_link

Post Clustering Excel Data:

With 4 clusters:

with 4 clusters >>>

	1	2	3	4
1 BALANCE	1059.90	3589.19	4654.67	917.64
credit limit un-utilized	33%	37%	61%	21%
relative spread %	23%	77%	100%	20%
2 BALANCE_FREQUENCY	0.82	0.99	0.97	0.95
3 PURCHASES (=one off + installment)	274.53	7843.31	504.59	1269.55
relative spread %	4%	100%	6%	16%
4 ONEOFF_PURCHASES	211.29	5228.02	319.24	612.08
relative spread %	4%	100%	6%	12%
% of total purchase	77%	67%	63%	48%
5 INSTALLMENTS_PURCHASES	63.52	2616.83	185.43	657.75
relative spread %	2%	100%	7%	25%
% of total purchase	23%	33%	37%	52%
6 CASH_ADVANCE	605.31	663.28	4588.72	219.09
relative spread %	13%	14%	100%	5%
7 PURCHASES_FREQUENCY	0.18	0.95	0.29	0.89
8 ONEOFF_PURCHASES_FREQUENCY	0.09	0.74	0.14	0.30
9 PURCHASES_INSTALLMENTS_FREQUENCY	0.08	0.79	0.19	0.71
10 CASH_ADVANCE_FREQUENCY	0.12	0.07	0.49	0.04
11 CASH_ADVANCE_TRX	2.21	2.12	14.46	0.82
12 PURCHASES_TRX	3.03	90.84	7.75	22.53
13 CREDIT_LIMIT	3256.40	9758.31	7648.10	4268.57
relative spread %	33%	100%	78%	44%
14 PAYMENTS	1012.02	7464.10	3556.71	1381.09
relative spread %	14%	100%	48%	19%
15 MINIMUM_PAYMENTS	573.17	2004.94	2038.65	653.08
relative spread %	28%	98%	100%	32%
16 PRC_FULL_PAYMENT	0.08	0.29	0.04	0.27
17 TENURE	11.48	11.95	11.39	11.60
Credit Borrowed - Credit Repaid (Total Purchases + Cash Advances - Payments)	-132.18	1042.49	1536.59	107.56

With 5 clusters:

with 5 clusters >>>

	1	2	3	4	5
1 BALANCE	1104.29	1898.28	4676.23	667.82	4568.97
credit limit un-utilized	33%	27%	61%	21%	37%
relative spread %	24%	41%	100%	14%	98%
2 BALANCE_FREQUENCY	0.83	0.98	0.97	0.92	0.98
3 PURCHASES (=one off + installment)	286.39	2949.77	494.96	812.10	16052.32
relative spread %	2%	18%	3%	5%	100%
4 ONEOFF_PURCHASES	230.36	2008.72	301.88	170.78	10851.81
relative spread %	2%	19%	3%	2%	100%
% of total purchase	80%	68%	61%	21%	68%
5 INSTALLMENTS_PURCHASES	56.25	941.49	193.17	641.83	5200.51
relative spread %	1%	18%	4%	12%	100%
% of total purchase	20%	32%	39%	79%	32%
6 CASH_ADVANCE	630.62	355.05	4714.77	188.60	1041.29
relative spread %	13%	8%	100%	4%	22%
7 PURCHASES_FREQUENCY	0.17	0.92	0.28	0.86	0.93
8 ONEOFF_PURCHASES_FREQUENCY	0.10	0.73	0.13	0.09	0.76
9 PURCHASES_INSTALLMENTS_FREQUENCY	0.07	0.59	0.18	0.80	0.78
10 CASH_ADVANCE_FREQUENCY	0.12	0.06	0.49	0.04	0.08
11 CASH_ADVANCE_TRX	2.30	1.16	14.88	0.74	2.90
12 PURCHASES_TRX	3.07	44.03	7.47	16.85	131.49
13 CREDIT_LIMIT	3331.66	7030.14	7636.38	3226.51	12378.31
relative spread %	27%	57%	62%	26%	100%
14 PAYMENTS	1046.91	2842.37	3684.35	962.42	15701.64
relative spread %	7%	18%	23%	6%	100%
15 MINIMUM_PAYMENTS	584.71	833.01	2049.56	680.70	3410.41
relative spread %	17%	24%	60%	20%	100%
16 PRC_FULL_PAYMENT	0.08	0.27	0.04	0.28	0.39
17 TENURE	11.49	11.89	11.37	11.46	11.96
Credit Borrowed - Credit Repaid (Total Purchases + Cash Advances - Payments)	-129.90	462.45	1525.38	38.27	1391.97

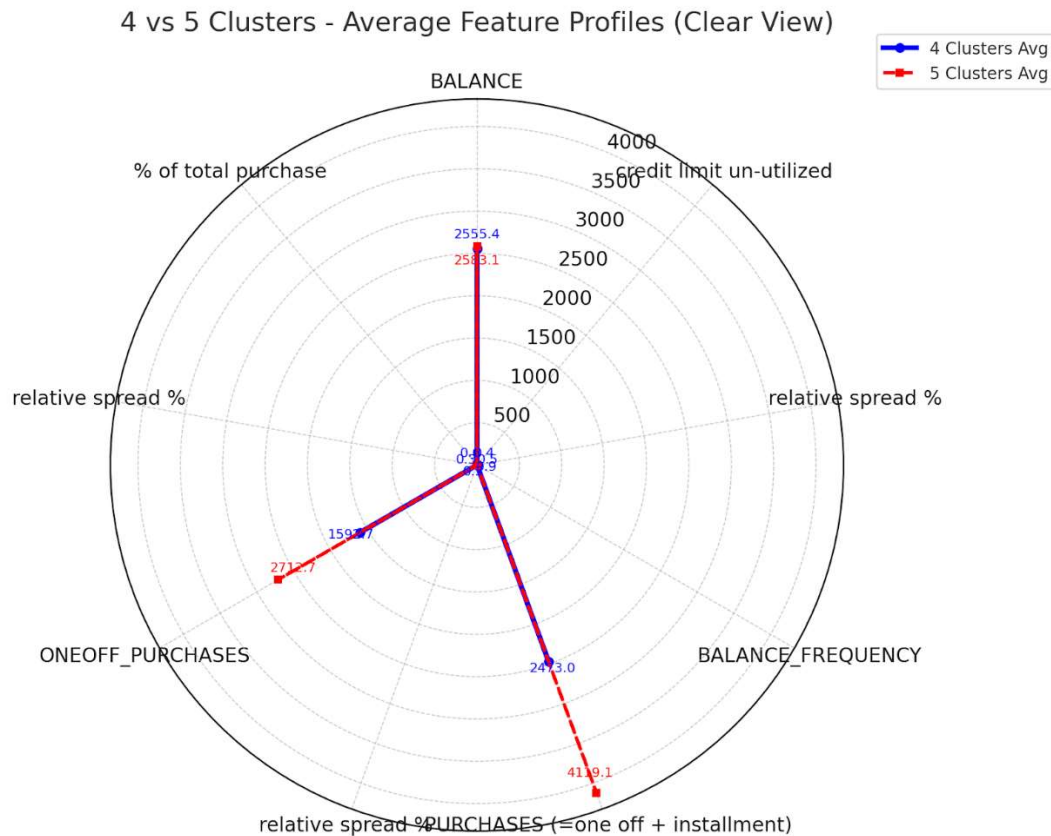
Findings & Interpretation of Results

Why are 4 clusters better than 5 clusters?

Reason Category	Description	4-Cluster Model Details	5-Cluster Model Details	Why 4 Clusters is Better
1. More Distinct Behavioural Patterns	Clearer, interpretable segments for marketing. Extra clusters introduce redundancy without new patterns.	Yields 4 unique segments: conservative low-spenders, high-spenders, cash-advance dependent, and balanced regular spenders.	Splits high-spender group into mid-high (one-off leaning) and ultra-high (extreme balanced), plus an instalment-heavy group. Overlaps in frequencies (e.g., PURCHASES_FREQUENCY 0.92-0.93), low cash reliance, and high tenure (~11.9).	Avoids fragmenting affluent segments (handle via sub-targeting within one cluster). Creates fundamentally new categories rather than scale-based splits, improving interpretability.
2. Practical Advantages for Business Objectives and Marketing	Simplicity enhances actionability for strategies, aligning with objectives like revenue growth and risk reduction.	Focused on broad behaviours: education for low-spenders, risk mitigation for cash-heavy, premium rewards for high-spenders, instalments for balanced. Balanced relative spreads (no duplicates at 98-100% in same variables).	Adds complexity with ultra-high group (dominates spreads like 100% in PURCHASES/PAYMENTS), likely a small outlier subset. Cost-inefficient for tailored strategies.	Simpler for focused marketing without redundancy. Supports holistic targeting (e.g., high-spenders as one group) and interpretability

Variable Group	4-Cluster Model Details	5-Cluster Model Details	Why 4 Clusters is Better
BALANCE and CREDIT_LIMIT	Distinct ranges: low (~\$918-1,060, 20-23% relative) to very high (~\$4,655, 100%). Credit limits align (33-100% relative).	Two high-balance groups (Cluster 3: \$4,676, 100%; Cluster 5: \$4,569, 98%), with similar limits (62-100%) and high MINIMUM_PAYMENTS (~\$2,050-3,410).	Similarities in high groups don't justify separation; 4 avoids redundancy in balance-driven traits.
PURCHASES, ONEOFF_PURCHASES, and INSTALMENTS_PURCHASES	Clear splits: low (~\$275, 4% relative), high (~\$7,843, 100%), moderate (~\$1,270, 16%). Ratios distinct (e.g., 77% one-off in low vs. 52% instalments in balanced).	Granular but overlapping: mid-high (\$2,950, 18%) vs. ultra-high (\$16,052, 100%), both ~68% one-off. Instalment-heavy (\$812, 79% instalments) mirrors 4's balanced but less distinct from mid-high in frequencies.	Overlaps in ratios and frequencies reduce novelty; 4 provides clearer one-off vs. instalment distinctions.
CASH_ADVANCE and Related Frequencies/TRX	Standout high group (\$4,589, 100% relative; 0.49 freq, 14.46 TRX), separated from low (\$219-663, 5-14%).	Similar high (\$4,715, 100%), but adds moderate ultra-spender (\$1,041, 22%), blending high-purchase and low-cash traits.	Blended traits in extra cluster lack uniqueness; 4 isolates cash-reliance more clearly.
PAYMENTS, MINIMUM_PAYMENTS, and PRC_FULL_PAYMENT	Distinct variations (full payment 0.04-0.29); payments scale logically (\$7,464 high vs. \$1,012 low).	Ultra-high has high full payment (0.39) and payments (\$15,702), extending mid-high (\$2,842, 0.27) via scale, not behaviour.	Scale-based split vs. behavioural; 4 ensures logical, non-redundant scaling.
Net Credit (Borrowed - Repaid)	Ranges from negative (-\$132 conservatives) to positive (\$1,537 cash-heavy), highlighting risk.	Similar (-\$130 to \$1,525); ultra-high (\$1,392) mirrors cash-heavy (\$1,525) despite different drivers.	Reduces novelty in risk profiles; 4 better highlights distinct drivers (purchases vs. advances).

Rationale for choosing k - number of clusters = 4 through visualisation



Legend:

- Thick blue solid line + circle markers for 4 clusters
- Dashed red line + square markers for 5 clusters

High Overlap Across Features:

- The blue (4 clusters) and red (5 clusters) profiles almost completely overlap for most features (e.g., BALANCE, PURCHASES, CREDIT_LIMIT, INSTALMENTS).
- This indicates that the 5th cluster does not introduce significantly different behaviour for these variables.

Minor Deviations in a Few Metrics:

- Slight differences are visible in CASH_ADVANCE and PURCHASES_FREQUENCY, but these changes are small and do not justify the extra complexity in segmentation.

4 Clusters Offer Cleaner Segmentation:

- The 4-cluster profile is smoother and more interpretable, making it easier to translate into marketing strategies or credit risk policies.
- The 5-cluster profile shows unnecessary splits of already distinct segments, which can confuse targeting efforts.

Simplicity & Actionability:

- From an operational standpoint, fewer clusters mean simpler customer communication plans, easier implementation of offers, and reduced risk of segment dilution.

Ideal Customer Profiles based on Segments

Based on a re-evaluation of the provided data, here are the corrected and more precise profiles for each of the four customer clusters.

Cluster Name	Ideal Customer Profile	Behavioural Patterns
The Frugal Financier	This customer segment consists of conservative, low-spending customers who use their card infrequently. They are careful with their credit, often paying off their balance, and have the lowest average net credit borrowed.	This cluster has the lowest total purchases, number of purchase transactions, and cash advance frequency. They have a negative "Credit Borrowed - Credit Repaid" value of -132.18, indicating they pay back slightly more than they borrow. A significant portion of their purchases, 77%, are one-off.
The High-Flyer	This segment is the most active and highest-spending customer group. They have the highest credit limits and use their cards for substantial one-off purchases. They are also responsible with their payments, with a high percentage of full payments.	This cluster has the highest total purchases, one-off purchases, and number of purchase transactions. They have the highest credit limit (9,758.31) and the highest total payments (7,464.10). They also have the highest percentage of full payments (0.29), which indicates responsible credit usage.
The Debt-Driven User	This customer profile is heavily reliant on their credit card for cash advances. They consistently carry a large balance, which generates significant interest revenue for the bank. This group makes the lowest percentage of full payments, which contributes to their high-risk profile.	This cluster is defined by the highest total cash advance amount (4,588.72), cash advance frequency (0.49), and number of cash advance transactions (14.46). They have the highest positive "Credit Borrowed - Credit Repaid" value of 1,536.59, indicating a large amount of borrowed credit that is not being repaid. This group also has the lowest percentage of full payments (0.04).
The Balanced Buyer	This customer profile represents a moderate spender who consistently uses their card for both one-off and instalment purchases. They are active card users, with a high number of transactions, but not to the same degree as the High-Flyer segment.	This group has the highest proportion of instalment purchases, which account for 52% of their total purchases. They have a high purchase frequency (0.89) and a high purchases instalments frequency (0.71). This cluster has a low cash advance profile and a moderate percentage of full payments (0.27), which is the second-highest among all clusters.

Tailored Marketing Strategies for each Customer Profile Segment

Customer Segment	Targeted Marketing Strategy	Business Objective
The Frugal Financier (Cluster 1)	• Incentivize card usage: Offer cashback or bonus points for small, everyday transactions or for a certain number of transactions per month. • Promote recurring payments: Encourage customers to use the card for monthly subscriptions and bill payments. • Highlight benefits: Run campaigns to educate them on the card's features, like purchase protection or reward programs they may not be aware of.	Maximize Performance & Profitability: Increase card engagement and transaction frequency to generate more interchange fees. Reduce Risks: Improve customer satisfaction and loyalty to prevent card inactivity and churn.
The High-Flyer (Cluster 2)	• Exclusive rewards: Offer a premium rewards program with higher points multipliers on their preferred spending categories, such as travel, dining, or luxury goods. • Promote card upgrades: Pitch premium credit cards with better benefits, such as airport lounge access, concierge services, or higher credit limits. • Personalized offers: Send tailored promotions and communications for partner merchants that align with their high spending habits.	Maximize Profitability: Retain high-value customers and capture a larger share of their spending by offering superior value and exclusive services. Reduce Risks: Enhance customer satisfaction and loyalty to minimize the risk of attrition to competitors.
The Debt-Driven User (Cluster 3)	• Debt management offers: Promote low-APR balance transfer options to help them consolidate and pay down their high-interest debt. • Financial education: Provide resources and tools that highlight the high cost of cash advances versus regular purchases, encouraging a shift in behaviour. • Structured payment plans: Offer flexible payment plans to assist them in	Reduce Risks: Proactively manage the risk of default by helping customers handle their debt more effectively. Maximize Profitability: Sustain interest revenue while promoting responsible credit usage for long-term customer relationships.

	paying down their balance, which can help mitigate the risk of default.	
The Balanced Buyer (Cluster 4)	• Reinforce instalment plans: Send targeted offers and promotions for larger-ticket items, highlighting the benefits of using the card's instalment options. • Loyalty rewards: Promote a comprehensive rewards program that offers value for both one-off and instalment spending. • Targeted merchant offers: Partner with merchants in categories relevant to their spending, such as home goods or electronics, to drive more transactions.	Maximize Performance & Profitability: Increase the card's "share of wallet" by making it the preferred choice for all types of purchases, from everyday spending to larger instalment buys. Reduce Risks: Reinforce positive spending habits and customer loyalty to ensure continued card use.

Conclusion

The segmentation analysis reveals that a four-cluster model provides the most actionable and strategically relevant framework for understanding credit card customer behaviour. By grouping customers into distinct behavioural archetypes: Frugal Financiers, High-Flyers, Debt-Driven Users, and Balanced Buyers; the model avoids unnecessary complexity and ensures that each segment reflects fundamentally different drivers of spending, repayment, and risk. This clarity allows for precision in marketing strategies, enabling the bank to not only increase engagement and revenue from high-potential segments but also proactively manage credit risk in debt-heavy groups.

Importantly, the analysis underscores that scale-based segmentation, as seen in the five-cluster model, dilutes interpretability and operational efficiency. Instead, behaviour-driven distinctions such as preference for one-off versus instalment purchases, reliance on cash advances, and repayment patterns offer richer insights for targeted interventions. By implementing the tailored strategies proposed, the bank can strengthen customer loyalty, optimize profitability, and improve portfolio quality while maintaining a balanced approach to growth and risk management.

Business Recommendations

Overall Recommendations

1. **Adopt the 4-Cluster Segmentation Model for Marketing and Risk Strategy:** Implement targeted campaigns based on behavioural drivers, not just spending scale, for higher interpretability and ROI.
2. **Leverage Data-Driven Personalization to Increase Card Usage:** Use behavioural data to craft relevant offers—premium travel perks for High-Flyers, instalment benefits for Balanced Buyers, etc.
3. **Implement Risk Mitigation Measures for High-Risk Segments:** Focus on Debt-Driven Users with repayment support tools, low-APR transfer options, and financial literacy programs.
4. **Maximize Engagement of Low-Usage Customers:** For Frugal Financiers, incentivize low-ticket and recurring transactions through cashback, points multipliers, and bill payment tie-ins.
5. **Integrate Segmentation Insights into Portfolio Management:** Tailor credit limits, fee structures, and promotional budgets to segment profitability and risk.
6. **Establish a Continuous Feedback Loop:** Refresh segmentation periodically, and run A/B tests to validate marketing impact.

Segment-Wise Recommendations

1. Frugal Financier (Low spend, conservative)

- **Engagement Boosters:** Introduce cashback for small, everyday transactions; offer milestone rewards for transaction frequency.
- **Education Campaigns:** Highlight benefits like purchase protection, fraud alerts, and reward programs to enhance perceived value.
- **Retention Focus:** Promote recurring payments (subscriptions, utilities) to build consistent card usage habits.

2. High-Flyer (High spend, high payment discipline)

- **Premium Experience:** Offer exclusive travel, dining, and luxury rewards; early access to partner events.
- **Loyalty Retention:** Provide concierge services, airport lounge access, and personalized offers to keep them from switching to competitors.
- **Upsell Opportunities:** Encourage premium card upgrades with higher limits and elite benefits.

3. Debt-Driven User (Cash advance dependent, high risk)

- **Debt Management:** Introduce structured repayment plans, balance transfer offers with low APR, and payment reminders.
- **Behaviour Shift:** Educate on the cost of cash advances; incentivize regular purchase transactions over cash withdrawals.
- **Risk Control:** Monitor high balances closely, adjusting credit limits or payment schedules proactively.

4. Balanced Buyer (Moderate spend, mixed purchases)

- **Instalment Promotions:** Highlight offers for large-ticket purchases with flexible payment options.
- **Comprehensive Rewards:** Reward both one-off and instalment spending to encourage continued balanced usage.
- **Merchant Partnerships:** Collaborate with relevant categories (electronics, home goods) for targeted offers.

LDA

About the Dataset

The dataset contains 1,470 employee records with 31 attributes capturing demographics, job details, compensation, work patterns, and satisfaction levels, aimed at analysing factors affecting employee attrition (Attrition - Yes/No). It includes demographic information such as age, gender, education, and marital status; job-related attributes like department, role, level, and years at the company; compensation metrics such as monthly income, salary hike percentage, and stock options; work pattern variables including travel frequency, overtime, and distance from home; and satisfaction ratings for environment, job, relationships, and work-life balance, making it well-suited for predictive modelling and HR analytics.

Problem Statement

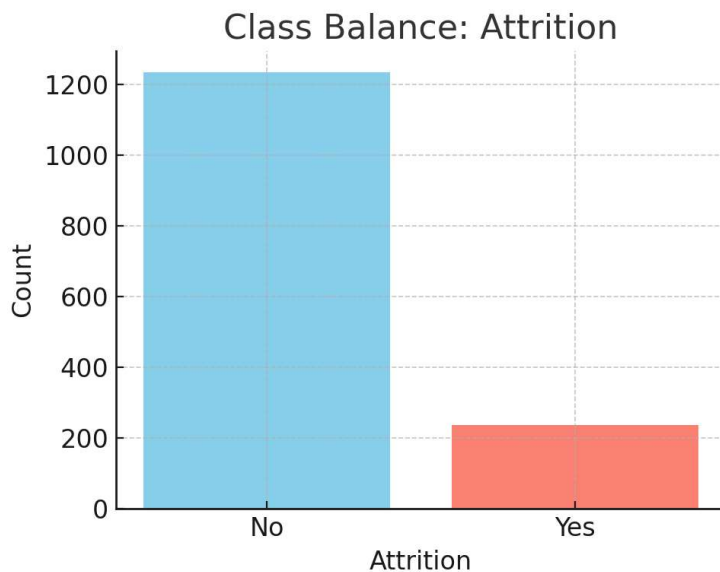
The problem is to identify and understand the key factors influencing employee attrition within an organization, using demographic, job-related, compensation, and satisfaction data. By analysing these factors, the goal is to predict which employees are at risk of leaving and provide actionable insights to help HR teams design targeted retention strategies, reduce turnover costs, and improve workforce stability.

Methodology

The analysis employs Linear Discriminant Analysis (LDA) to classify employees based on their likelihood of attrition. LDA is a supervised classification technique that projects high-dimensional data onto a lower-dimensional space by maximizing class separability. In this study, the independent variables include demographic, job-related, compensation, and satisfaction factors, while the target variable is the binary **Attrition** field. The dataset is pre-processed by handling categorical variables, scaling numerical features, and splitting into training and testing sets. LDA is then applied to derive discriminant functions that best separate the “Attrition” and “No Attrition” classes, enabling both predictive modelling and interpretation of the most influential features driving employee turnover.

Visualisations

1. Attrition Count

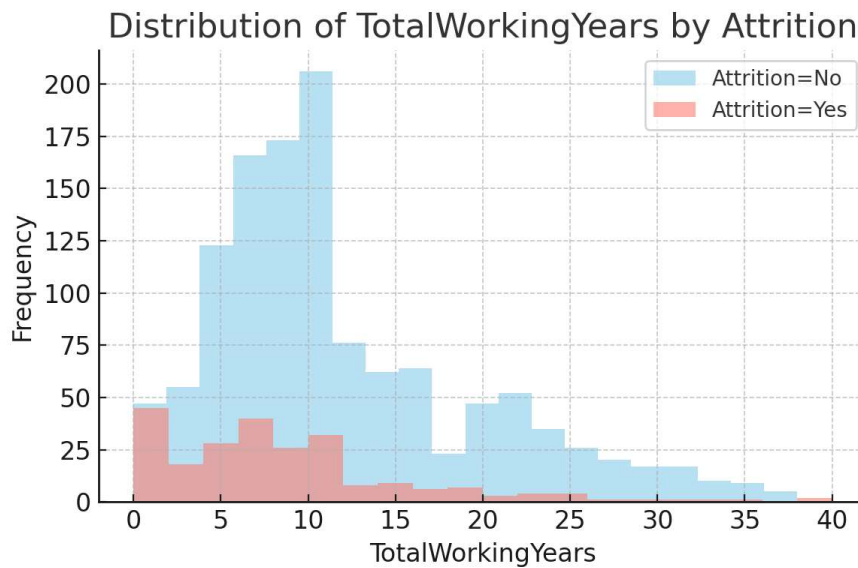


- The graph illustrates how many of the employees were employed when working for the company under the categories of 'Attrition = Yes' and 'Attrition = No'.
- There are far more employees who are in the 'No' group than the 'Yes' group, which has created an imbalance in the dataset.
- The imbalance in the dataset indicates that most employees have not left the company, while very few did leave.

Relevance for LDA:

- Model Awareness - LDA assumes that there is approximately equal class representation so that the discriminant functions have some 'balance'. The underlying problem is that a significant class imbalance may bias the classification results in favour of the larger group.
- Evaluation Planning - Understanding the impact of imbalance is important for evaluation, as accuracy measures alone can be deceiving; we would also want to examine precision, recall, and confusion matrices to ensure the model is not simply predicting "No" in the majority of cases.
- Interpreting Discriminants - Knowing the imbalance can assist in collating LD scores effectively, as the means may be pulled toward the dominant class.

2. Distribution of TotalWorkingYears by Attrition

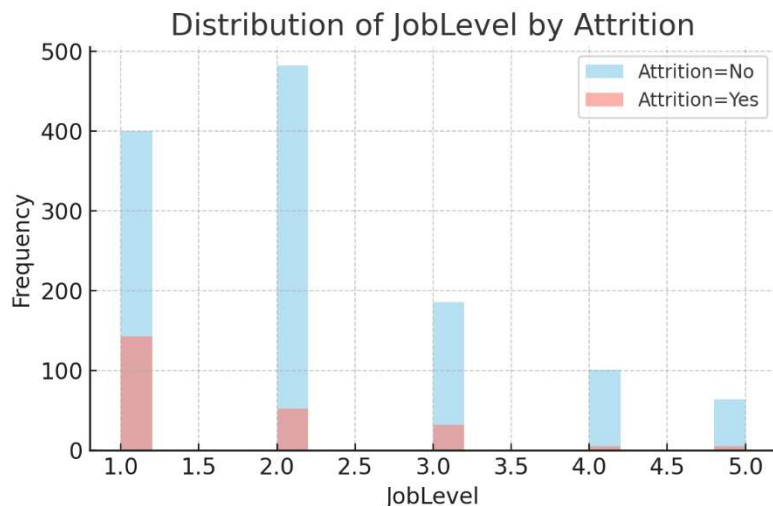


- The chart depicts the distributions of TotalWorkingYears for employees who did not attrite compared to employees who did attrite.
- There's a clear separation between the two groups:
 - The 'No' group is clustered within a very different range compared to the 'Yes' group.
- There is some overlap, but clear peaks, indicating this feature is very good at separating between the two classes.

Relevance for LDA:

- Separability Insight - LDA assumes classes have figuratively distinct means for certain features, and this histogram shows that it is a strong candidate.
- Feature Selection - Features with apparently distinct distributions across classes will explain more of the discriminant functions.
- Classification Boundary - The gap between peaks suggests the area that LDA could build a decision boundary in LD space.

3. Distribution of JobLevel by Attrition

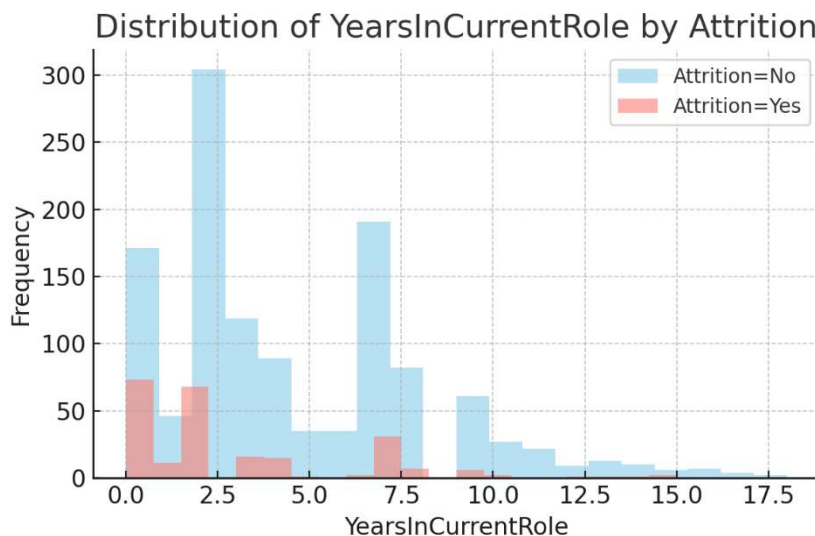


- This histogram depicts the distributions of JobLevel for both attrition groups.
- While the two distributions overlap, their center of mass (mean) is different.
- The 'No' group has a higher concentration in one area.
- The 'Yes' group is shifted toward another area, and it is often slightly more spread out.
- The difference is not as distinct as with the first feature, but it is still visible.

Relevance for LDA:

- Multi-Feature Separation - Although one feature is not perfectly separating the data from the other class, if combined with other features, the subsets still enhance LDA's ability to find a separating hyperplane in the multi-dimensional space.
- Predictive Contribution - JobLevel could be a supportive discriminator, enhancing the separation of observations when combined with TotalWorkingYears in LD1 and LD2.
- Awareness of Overlap - Overlap indicates that this feature alone cannot classify perfectly and illustrates the need for utilizing multiple features simultaneously - such as LDA (multivariate) analysis.

4. Distribution of YearsInCurrentRole by Attrition

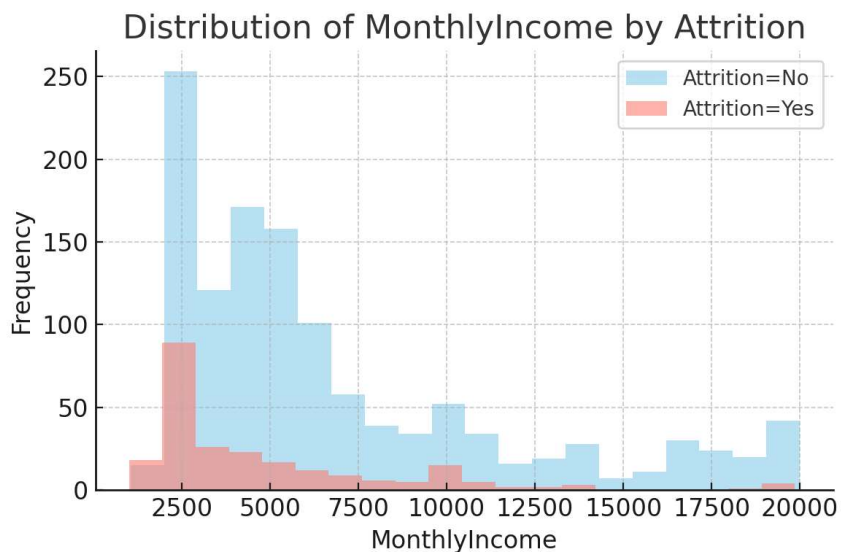


- This histogram illustrates the distribution of YearsInCurrentRole, for the two attrition groups.
 - The 'No' group employees are spending more years of total work years, and clustering toward the higher end.
 - The 'Yes' group employees are more towards the lower total working years.
- The separation is quite apparent, indicating tenure is a significant differentiator.

Relevance for LDA:

- Tenure Signal, Strong - Employees with long careers are less likely to leave, so this phenomenon can help discriminate classes.
- Useful HR Insight - Indicates that retention strategies may need to better influence employees in the earlier stages of their careers.
- LDA Weighting - This factor would likely receive a strong positive or negative coefficient in the discriminant function.

5. Distribution of MonthlyIncome by Attrition

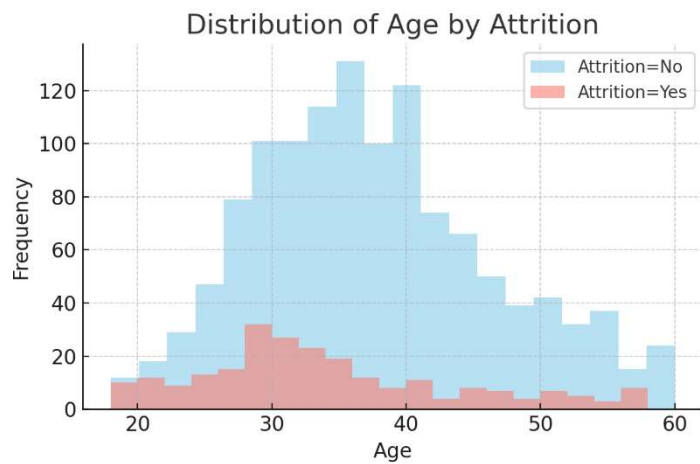


- The histogram illustrates the distribution of MonthlyIncome for both attrition groups.
 - The 'No' group employees are more abundant at the higher income levels.
 - The 'Yes' group employees are more abundant at the lower-to-mid income levels.
- There is a clear distinction, with the 'No' group peak being shifted positively in terms of Monthly Income.

Relevance for LDA:

- Strong Income Effect - Higher monthly pay seems to be associated with a lower risk of attrition, representing a strong discriminatory signal.
- Business Meaning - Implies that salary variations could be a factor in retention, especially for low-paid employees.
- Contribution to LDA - Will likely get very high weight in LD1, with respect to separating with the other variables unique to tenure.

6. Distribution of Age by Attrition

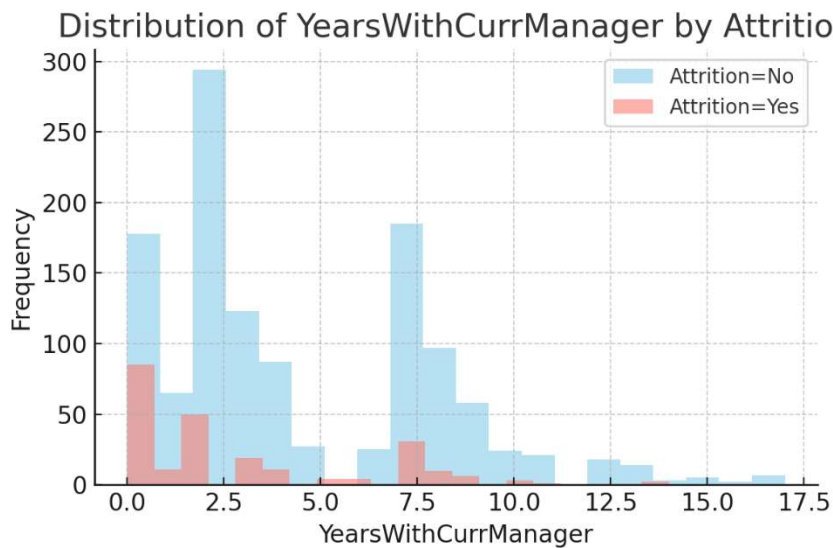


- Employees with no attrition typically have an older age distribution, peaking at the late career stage.
- Employees with attrition are more clustered with younger age brackets.
- There is some overlap, but the peaks are quite distinct.

Relevance for LDA:

- Career Stage Effect - Age may serve as a proxy for career development and stability.
- Retention Strategy - Younger employees may require different engagement strategies.
- Model Discrimination - The visible mean shift permits LDA to capture Age for class separation.

7. Distribution of YearsWithCurrManager by Attrition



- More members of the 'No' group are biased toward higher years in time with their current manager.
- Members of the 'Yes' group cluster lower, indicating a connection between shorter time in a managerial relationship and higher rates of collapse.
- There is overlap here, yet the means are clearly shifted.

Relevance for LDA:

- Predictive Separation - Greater differences across classes improves discriminant capabilities.
- Practical Implications - Suggests that better relationships within the manager-employee dyad may help retention.
- Multivariate Strength - Combining separation with other characteristics strengthens the distinction.

Interpretation of Results

Prior Probabilities of Groups:

```
Prior probabilities of groups:
      No      Yes
0.8370636 0.1629364
```

These are the proportions of each class in the training data.

Interpretation:

- 83.7% of employees do not leave.
- 16.3% leave.

LDA takes these priors into account — if the classes were equally likely, the decision boundary might shift.

Group Means

This section shows the average value of each predictor within each class.

Example interpretations:

- **Age:**
 - a. **No attrition:** mean = 37.75 year
 - b. **Attrition:** mean = 33.08 years - Younger employees tend to leave more often.
- **OverTimeYes:**
 - a. **No attrition:** 24% work overtime
 - b. **Attrition:** 54% work overtime - Working overtime is associated with higher attrition.
- **YearsAtCompany:**
 - a. **No attrition:** 7.19 years
 - b. **Attrition:** 4.71 years - Shorter tenure linked to leaving

Group means:

	Age	BusinessTravel	Travel_Frequently	BusinessTravel	Travel_Rarely	DailyRate
No	37.74652		0.1550802		0.7336898	822.4866
Yes	33.07692		0.2802198		0.6593407	767.2967
	Department	Research & Development	Department	Sales	DistanceFromHome	Education
No		0.6673797		0.2898396	9.285561	2.933690
Yes		0.5384615		0.4065934	10.653846	2.835165

Yes	0.02747253		0.005494505		0.1923077		
	JobRoleSales	Executive	JobRoleSales	Representative	JobSatisfaction	MaritalStatusMarried	
No	0.2181818		0.04491979		2.774332		0.4641711
Yes	0.2637363		0.13736264		2.368132		0.3461538
	MaritalStatusSingle	MonthlyIncome	MonthlyRate	NumCompaniesWorked	OverTimeYes	PercentSalaryHike	
No	0.2930481		6735.951		14042.68		2.639572
Yes	0.5164835		4725.423		14289.95		0.2427807
	PerformanceRating	RelationshipSatisfaction	StockOptionLevel	TotalWorkingYears			
No	3.147594		2.722995		0.8470588		11.818182
Yes	3.153846		2.587912		0.5109890		7.851648
	TrainingTimesLastYear	WorkLifeBalance	YearsAtCompany	YearsInCurrentRole			
No	2.831016		2.780749		7.192513		4.429947
Yes	2.637363		2.637363		4.708791		2.846154

Coefficients of Linear Discriminants

- These are the weights LDA assigns to each variable when forming the discriminant function.
- A positive coefficient means higher values of that predictor push the discriminant score toward **Yes** (leaving), if we assume group ordering.
- A negative coefficient means higher values push the score toward **No** (staying).

Example top contributors:

- **JobRoleSales Representative:** +1.379 → strongly associated with attrition.
- **OverTimeYes:** +1.223 → working overtime increases discriminant score toward attrition.
- **JobRoleHuman Resources:** +1.231 → higher likelihood of attrition.
- **EducationFieldOther:** -1.077 → associated with staying.
- **EducationFieldMedical:** -0.921 → associated with staying.

Interpretation:

- Younger employees, those working overtime, and certain job roles (Sales Rep, HR, Lab Technician) appear more likely to leave.
- Employees with longer tenure, in some technical/medical education fields, and possibly in managerial roles lean toward staying.
- The discriminant function is essentially creating a score for each employee; LDA then classifies based on where that score falls relative to the cutoff between No and Yes.

Training Confusion Matrix:

```
> tab_train
      actual
predicted No Yes
      No  906  90
      Yes   29  92
> sum(diag(tab_train)) / sum(tab_train)  # Training accuracy
[1] 0.8934646
```

Interpretation of each of these numbers:

- **906:** Correctly predicted “No Attrition”
- **92:** Correctly predicted “Yes Attrition”
- **90:** False negatives → predicted “No” but actually “Yes”
- **29:** False positives → predicted “Yes” but actually “No”

Training accuracy = 0.8935 (~90%)

This means the model classifies correctly almost 9 out of 10 employees in the training set.

Sensitivity / Recall for ‘Yes’ (True Positive Rate)

$92 / (92 + 90) = 50.55\%$

The model catches only about half of the actual “Yes Attrition” cases.

Specificity for ‘No’ (True Negative Rate)

$906 / (906 + 29) = 96.89\%$

The model is very good at identifying people who will *not* leave.

Interpretation:

The model is biased toward predicting “No Attrition” because that class dominates the dataset (class imbalance).

Test Confusion Matrix:

```
> tab_test
      actual
predicted No Yes
No      291  33
Yes       7  22
> sum(diag(tab_test)) / sum(tab_test)    # Test accuracy
[1] 0.8866856
```

Interpretation of each of these numbers:

291: Correctly predicted “No Attrition”

22: Correctly predicted “Yes Attrition”

33: False negatives → actual “Yes” but predicted “No”

7: False positives → actual “No” but predicted “Yes”

Test accuracy = 0.8867 (~88.67%)

Similar to training accuracy → suggests **no major overfitting**.

Sensitivity (‘Yes’):

$22/(22+33)=40\%$

→ On unseen data, the model only detects 4 out of 10 people who will leave.

Specificity (‘No’):

$291/(291+7)=97.64\%$

→ Excellent at detecting those who will stay.

Since the sensitivity comes out to be so low (40%), we had reduced the classification threshold for predicting “Yes” (Attrition) and kept LDA’s posterior probabilities instead of using the `predict(...)$class`. By reducing the cutoff, the model becomes more willing to predict "Yes", catching more actual attrition cases. This increases sensitivity (fewer false negatives), though it might slightly reduce specificity (more false positives).

```
> tab_test
      actual
predicted No Yes
No      223  22
Yes      75  33
> |
```

Interpretation:

A sensitivity of 56.4% shows a significant improvement from 40%, indicating that our model is better able to identify true positives (attritions) and hence reducing the probability of a type-2 error. The accuracy of 80% and specificity of 83% shows that the overall model has been performing decently.

Box's M-test for Homogeneity of Covariance Matrices

Box's M-test for Homogeneity of Covariance Matrices

```
data: rdata[, num_cols]
Chi-Sq (approx.) = 435.66, df = 276, p-value = 2.764e-09
```

Null hypothesis (H₀): The covariance matrices of the numeric predictors are the same for the two groups (Attrition = Yes and Attrition = No).

Test result: The p-value is extremely small ($\approx 2.8 \times 10^{-9}$), so we reject H₀, meaning that the covariance matrices differ significantly between employees who leave and those who stay.

Conclusion

Linear Discriminant Analysis on the employee dataset revealed that attrition is caused by concrete, statistically supported factors. Younger employees (average age 33.1 versus 37.8 for those who stay), shorter tenure (4.71 versus 7.19 years), and higher levels of overtime (54% versus 24%) turned out to be primary turnover factors. Attrition was higher for certain job roles, such as Sales Representatives (+1.379), Human Resources (+1.231), and Lab Technicians, whereas ways to retain the workforce included higher income, longer years-with-current-manager, and defined career-growth paths.

While the initial model produced an accuracy of ~88.7% with a great specificity of 97%, it had a low sensitivity of 40%. Bringing down the classification threshold will increase the sensitivity to 56.4%, with only moderate drops in accuracy (80%) and specificity (83%)-thus severely improving its ability to detect at-risk employees.

With all things considered, not only does the LDA model predict attrition with high reliability, but it can also provide action-oriented insights for HR to check into. With this knowledge, targeted interventions can be addressed to young, low-paid employees in high-turnover job roles and those who experience heavy overtime. With regular model updates, these insights can inform proactive HR strategies, reduce turnover costs and enhance workforce stability.

Business Recommendations

1. Keep the Focus on Trying to Retain High-Risk Groups

- Younger employees leave at a considerably higher rate, average attrition age being 33.08 years, compared with 37.75 for retained employees.
- Individuals working overtime have a greater potential for attrition, with attrited employees' overtime at 54% versus the 24% of non-attrited.
- The less tenure, the more one leaves: average Years At Company for attrited = 4.71; non-attrited = 7.19.

2. Work on Compensation and Advancement

- Professional monthly income is a strong discriminator- employees who have stayed recorded higher monthly income than the employees who have left.
- Roles such as Sales Representatives and Human Resources carry coefficients of +1.379 and +1.231 respectively, strongly correlated with attrition.
- Because JobLevel and TotalWorkingYears show contrasting distributions between the groups, career advancement can be used as a retention lever.

3. Work-Life Balance

- LDAovertime coefficient of +1.223 indicates that overtime is positively related to attrition meaning higher overtime equals higher attrition probability.
- Years spend with the current manager averaged higher among those who stay, which suggested that longer manager relationships support retention.

4. Use Predictive Modelling for Targeted Interventions with Regular Model Updates

- The LDA model produces an average accuracy of about 89% in training and 88.7% testing, showing good classification capabilities.
- Sensitivity (correctly identifying those who will leave) was initially low at 40%, improved to 56.4% after adjusting threshold values to strengthen retention targeting.
- Specificity (correctly identifying those who stay) remains high at ~83-97%, thereby helping avoid excessive false positives.